COMPOSING SOUNDSCAPES FROM A SINGLE FIELD RECORDING

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ABSTRACT

This paper will discuss the multiple uses of a single original field recording in relation to the author's own work, with inspiration drawn from composer Joanna Bailie's works. Bailie views the prolongation of frequencies in field recordings as a way of "freezing" a section of time, to explore the essences of the frequencies found within a particular point in time - turning the moment into a decipherable stretch of time for the mind. This elongation also allows the exploration of emotion and musicality of a moment, by exposing hidden harmonies and rhythms. The paper will discuss methods of composing with a single field recording (time-stretching, layering, and the use of effects and panning), through examining two pieces from the author - *Corridor* [2018] and *The Insect Apocalypse* [2018] - and how such a process, with its combination of artistic and technological interventions, can also create pieces that mediate on the ecological stresses found within the original field recording.

1. INTRODUCTION

My start in field recordings began in 2015, as I began to explore ways I could incorporate my experiences in a place into compositions. While I was initially viewing field recordings as either recording-as-documentation or gathering sonic material, with the focus being on capturing impartial and neutral artifacts of soundscapes, I became increasingly aware of the role of field recordings as a form of personal expression. As I worked more deeply with field recordings, I became increasingly interested in the narrative side of my process - for instance, the reasons why I chose a particular place to visit and record, the sounds I was choosing to record and how I was using and arranging sounds to create the composition.

I found that the more I listened, the more I noticed the depths of complexity in the recorded sounds. I became interested in how manipulating sounds through the use of effects, time-stretching, and playing with time and asynchronicity, as though the layers were like multiple waves hitting and bumping into each other. I found I could layer these manipulated sounds with the original recording as a way of reflecting my experiences in the landscape.

This paper will be split into two broad sections – firstly, a discussion of composition using single-source field recordings, with a focus on UK composer Joanna Bailie, and secondly, a discussion of two of my works inspired by Bailie's approach - *Corridor* (2018) and *The Insect Apocalypse* (2018).

2. FIELD RECORDINGS AS NARRATIVE – A BRIEF OVERVIEW

Field recordings have been perceived as recording-as-documentation or as a process of gathering sonic material, with the focus being on capturing impartial and neutral artifacts of soundscapes. The narratives that exist within the recordist, however, have been generally less acknowledged. Isobel Anderson and Tullis Rennie noted in their article, *Thoughts in the Field: Self-reflexive narrative in field recording*, where they discussed notions around narrative in field recording: "These narrative details should certainly not automatically be silenced, repressed, or re-dated, which are common conventions within the practice. Instead, these insights can become some of the most interesting and creative elements of field recordings, both strengthening the field recording artist's understanding of their practice and providing greater potential engagement for listeners" (Anderson and Rennie, 2016).

There have been several composers who have explored field recordings as a form of self-reflexive narrative. Arguably, one of the earliest practitioners was French composer Luc Ferrari. Ferrari's first foray, *Presque rien, ou le lever du jour au bord de la mer* (Almost nothing, or daybreak at the seashore), recorded in Vela Luka, Croatia and released in 1970, was a demonstration of a field recording being used both as a part of composition and auditory memory making. While in Vela Luka, Ferrari spent several weeks exploring the village and surrounds with his partner, Brunhild Meyer Ferrari. As they explored the area, Ferrari became increasingly drawn to the sonic environment and his responses to what was happening, and recorded for several mornings. As noted by Lawrence English in his essay about the work, "In *Presque Rien No. 1,* Ferrari perceived that his listening, as an affective and agentive performance within a given horizon, was not absolute or ongoing, but rather highly

selective in attentiveness and temporality" (English, 2017).

Another example is seminal soundscape artist Hildegard Westerkamp, with works such as *Kits Beach Soundwalk* - where she explores her affinity with Vancouver's Kits Beach through a combination of field recordings and Westerkamp's spoken contemplations, which demonstrates, to the listener, Westerkamp's active role in capturing and reflecting on the soundscape. One point of particular interest with *Kits Beach Soundwalk* is how Westerkamp openly sign-posts to listeners her manipulations in the studio, such as informing audience she is diminishing the sound of traffic to amplify the sound of the ocean because the view is "…beautiful. In fact, it is spectacular" (Anderson and Rennie, 2016).

In a similar vein to Westerkamp, Janet Cardiff also establishes herself within soundscape recordings through her series of soundwalk compositions, where she narrates over a composed soundscape. In works such as *A Large Slow River*, Cardiff uses Lake Ontario as a setting for discussing memory and time. For instance, in one passage Cardiff narrates over the sound of crashing waves: "I'm at a beach on Lake Huron, my toes squishing into the mud…jumping off my father's wet shoulders into the water. Now I'm at another beach, it's night, the sound of the waves coming in through the screen windows" (Cardiff, accessed September 1 2019).

Manipulating the temporalities within recordings themselves is another way in which composers can use field recordings as self-reflexive narration. A prominent example is Barry Truax, who explores temporality and narrative through the elongation of field recordings. Truax explains his technique as a way of revealing "...the inner harmonics within familiar sound signals...My intent was to use the suspension of these sounds to give listeners the space in which to explore their own inner associations..." (Truax, 1992).

One example of this is *Pacific* (1990), with the first movement *Ocean* based on a recording of waves hitting a sandy beach. *Ocean* begins with the waves played at normal speed, but gradually stretched until it gives the listener the sensation of a "gradual submersion beneath the waves as their inner timbre is revealed…" (Truax, 1992). The process of submersion, as Truax explains, "…is accompanied by the loss of the rhythmic pattern of the waves, which further heightens the illusion of entering a normally hidden realm" (Truax, 1992).

The idea of slowing recordings as a way of revealing hidden sounds is taken further by composer Joanna Bailie.

3. JOANNA BAILIE

Joanna Bailie is a composer and musician originally from London, but now based in Berlin. Her recent works include chamber music and installation and are characterised by her use of field recordings alongside acoustic instruments. Bailie's passion lies in exploring the ways in which elements of the 'real' world can be recorded, then manipulated and transcribed for acoustic instruments to be performed alongside the processed field recording itself. For Bailie, this process allows a "…fashioning [of] a more complex relationship [between sound and listener] that lies beyond the idea of a simple immersion into a pseudo-reality" (Bailie, 2014).

Bailie describes one of her first experiences with field recordings and what initially drew her towards her approach:

> *I remember once sitting at a bus stop recording the cars go by and at one point there was a shift in my perception, perhaps due to my level of concentration or the effect of amplification. I genuinely had the impression that the cars were driving past a particular times, speeds and volumes for precisely my own pleasure, that it sounded good and that it was indeed music* (Power, 2014).

Similar to Truax, Bailie employs time manipulation techniques where she elongates sounds to, in her words, "…allow for the exploration of emotion and musicality of a moment, by exposing hidden harmonies and rhythms". (Bailie, 2014). The resultant composition is one that is filled with technological and artistic intervention by the artist, rather than it being a reflection of the original subject of the sound recording. Bailie explains, "A possible approach may be to actively look for music when making field recordings or to tease the music out of the recording by some kind of manipulation. For me it's a question of an appealing dramaturgy that might itself suggest a compositional strategy, a certain (fortuitous) balance of elements and more often than not, pitch content, whether it comes in the shape of music in public spaces, car horns or airplane drones."

Bailie terms the process of elongating sounds as 'freezing', and describes the concept while discussing the processes behind one of her pieces, Artificial Environment No. 3:

> *Artificial Environment No. 3 is a place where time is stopped and started in an aperiodic and therefore unpredictable manner. As frustrating as this must be for its inhabitants, it has a rather curious effect on sound. At the moment of freezing, a split second of sonic resonance is prolonged until life resumes again from the point where it was broken off. Even stranger is the effect that these freezes have on the general ambience. Sounds start to behave like an inappropriately sentimental soundtrack. A bell, a passing car, or a fragment of birdsong are given an air of gravitas and occasional nostalgia simply by being taken out of the continuum of everyday life and stretched into a chord* (Power, 2014).

For Bailie, the process of 'freezing' becomes a method of revealing aspects about the world that might

otherwise escape perception. She notes that by transforming an 'instant' – that is, an event of such short duration that it becomes imperceptible – into a longer 'moment', it allows the event to inhabit the mind and become accessible to memory. Bailie elaborates on this in an article about her process: "…it might lead us to think about some of the emotions that are evoked through the prolongation in time of a split-second…" (Bailie, 2014). Often Bailie combines these elongations together intuitively, following the flow of a narrative over any temporal hierarchy.

4. WORK 1: CORRIDOR

I found Bailie's work particularly inspiring because of her desire to explore the emotionality behind sound through probing the sonic nuances that lie within a recording. There is a powerful evocativeness in Bailie's work - from its source from real, occurring events to the strange, dissonant and surprisingly melodies and rhythms that are teased out from being 'frozen'. I have also become increasingly preoccupied with climate change and wanted to use field recordings as a way to discuss environmental concerns.

One piece I wrote that was inspired by Bailie's approach was *Corridor*[1]. *Corridor* is a piece exploring the use of a single sound source to construct a soundscape - as though the sound is mediating on itself. This piece is also about movement, loss and attempting to preserve what was left.

The original field recording was made in an underpass in Neerabup National Park, in the northern suburbs of Perth, Western Australia. Neerabup is an important stretch of bush for two main reasons - firstly, it acts as a wildlife corridor for native wildlife (particularly the many native birds), connecting two important wetland areas in Lake Joondalup and Loch McNess. Secondly, it incorporates and protects part of an ancient Aboriginal migration route, which is now a walking trail called the Yaberoo Budjara Heritage Trail.

When composing this piece, I wanted to explore the juxtapositioning of an historically and ecologically important area within a busy transport network. This particular section of Neerabup National Park is bisected by Neerabup Road, a busy road which provides access to the Neerabup industrial area. Bordering Neerabup National Park are two main roads – the Mitchell Freeway, and Wanneroo Road – plus the train line.

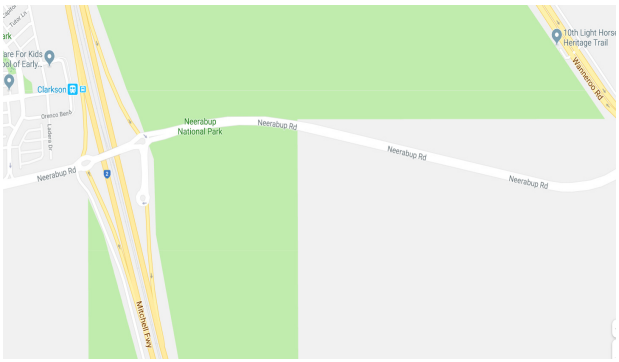


Figure 1. A map of the area where the recording was made, showing the roads that bisect and border Neerabup National Park (Map data: Google).

The initial recording was made during the afternoon, when traffic was busy. In the original recording, the sound of birds and swaying trees can be heard in the background, with the sounds of vehicles passing threatening to overtake these sounds. I felt this recording was a microcosm of the continual tensions between preserving and protecting the ecosystem with humanity's need for development, and wanted to explore this further in the composition.

One aspect I wanted to accentuate in the piece was the deep drones of the vehicles as they drove over the overpass, so I slowed and pitch-shifted the original recording in Adobe Audition by various degrees (800%, 600%, 400% and 200%). I also added a doppler effect (also in Adobe Audition) to one of these versions to replicate the sound of wings flapping, as a way of representing the movement of birds through the wildlife corridor. These various versions were then layered with the original in Ableton Live. The layering of tracks was my way of playing with temporality – I viewed the building of the composition as though they were waves tumbling through and onto each other as they approached land, each in their own as they individually navigate the geography of the sonic shoreline.

The final composition consisted of eight layers, with the following effects:

Layer 1: Original recording
Layer 2: Original recording + delay
Layer 3: Excerpt from original of cars passing, slowed 800%
Layer 4: Layer 3 + resonator
Layer 5: Excerpt from original of cars passing, slowed 600%
Layer 6: Excerpt from original of cars passing, slowed 400%
Layer 7: Excerpt from original of cars passing, slowed 200%
Layer 8: Layer 7 with doppler effect

[1] Full piece can be heard here: https://samarobryn.bandcamp.com/track/corridor-disquiet-0348

5. WORK 2: THE INSECT APOCALYPSE

The Insect Apocalypse is a series of two pieces, based on a recording of crickets outside the front of my house. The title of my piece references an article called 'The Insect Apocalypse Is Here' where insect populations around the world have dropped dramatically in recent years due to human-related reasons (Jarvis, 2018). I was inspired to ponder how an 'apocalypse' might sound and feel like from an insect's perspective. *The Insect Apocalypse* is split into two sections:

The Insect Apocalypse: a piece imagining how the 'apocalypse' is perceived as it is happening, and therefore features mechanical-like sounds as a way of representing human machinery and interference.

Heart Of The Apocalypse: a piece that imagines how the internal experience of the 'apocalypse' might be experienced by an insect as it slowly dies.

The first track of the series, *The Insect Apocalypse*[2], features a field recording of crickets in the foreground with sounds of passing traffic in the background. I wanted to create a mechanical droning sound to the piece as an ominous backdrop, so I pitch-shifted and slowed the original recording in Adobe Audition by 1200% and 800%, and added a doppler effect to one of the versions. The rest of the composition was completed in Ableton Live, where I added delay, panning, resonator and distortion effects to further invoke a threatening soundscape. I was particularly drawn to using panning effects and arranging sounds in an asynchronous manner (i.e. playing two versions of the same sound, 30 seconds apart) to create an immersive and dissonant experience for the listener.

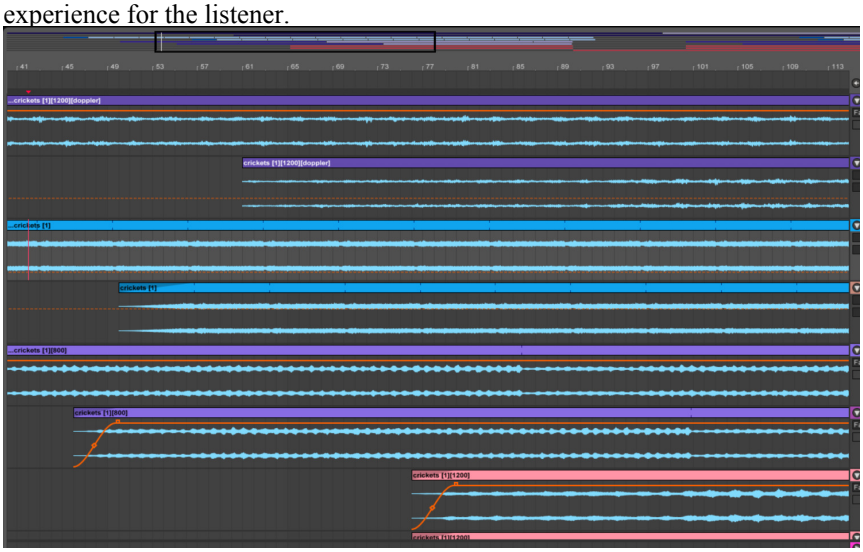


Figure 2. Screenshot of the construction of *The Insect Apocalypse* in Ableton Live, showing some of the tracks being layered in an asynchronous manner. This screenshot is taken from the beginning of the piece, with the first six layers.

The final composition consisted of nine tracks:

Layer 1: Original recording slowed 1200%, with doppler effect
Layer 2: Original recording slowed 1200%, with doppler effect, Moscow resonator, distortion and Wanderer delay effect
Layer 3: Original recording with Around The Head panning effect
Layer 4: Original recording with Around The Head panning effect and Low Tone Flutter grain delay
Layer 5: Original recording slowed 800%, panned 90-degrees left with Sennheiser AMBEO Orbit (for binaural panning)
Layer 6: Original recording slowed 800%, panned 90-degrees right with Sennheiser AMBEO Orbit
Layer 7: Original recording slowed 1200%, with Mirage bit reduction and panned 45-degrees right with Sennheiser AMBEO Orbit
Layer 8: Original recording slowed 1200%, with Swirling phaser and panned 45-degrees left with Sennheiser AMBEO Orbit
Layer 9: Original recording slowed 1200%, with Moscow resonator and reverb.

The second piece in the series, *Heart Of The Apocalypse*[2], continues in same vein but with a singular "frozen" cricket chirp recording taken from the original field recording of the crickets in my front yard. For this piece, I was particularly drawn to Bailie's ideas behind her *Artificial Environment* series, where she noted how elongation exposed the hidden complexities within an otherwise short sound.

The original cricket chirp lasted less than a second, but using Adobe Audition, I was eventually able to stretch the sound to nearly 9 minutes. After this pitch-shifting, I experimented with the graphic equalizer option to cut out specific frequency bands and was struck by the complexity of the revealed harmonies. Organising the resultant versions of the elongated field recording was reminiscent to arranging an eerie choir.

The final composition consisted of seven layers, with the following effects:
Layer 1: Time-stretched recording of cricket chirp (used as the foundation or "base" of the composition) with Around The Head panning effect
Layer 2: "Base" recording with

[2] Full piece can be heard here: https://samarobryn.bandcamp.com/track/the-insect-apocalypse

[3] Full piece can be heard here: https://samarobryn.bandcamp.com/track/heart-of-the-apocalypse



Figure 3. Screenshot of the construction of *The Heart Of The Apocalypse* in Ableton Live, showing how some of the variations of the field recording were layered. This screenshot starts at beginning of the piece, with the first six layers.

Ring frequency shifter effect
Layer 3: "Base" recording filtered at 250hz
Layer 4: "Base" recording filtered at 500-710hz and panned 45-degrees left with Sennheiser AMBEO Orbit
Layer 5: "Base" recording filtered at 500-1000hz and panned and panned 45-degrees right with Sennheiser AMBEO Orbit
Layer 6: "Base" recording filtered at 250hz with Five grain delay, distortion and Around The Head panning effect
Layer 7: "Base" recording with Frozen Build Up reverb effect, distortion and Around The Head panning effect

6. CONCLUSION

Basing a composition on a single field recording can allow a deep engagement with the recording itself, particularly when combined with technological and artistic intervention by the artist. Methods such as time-shifting allows an artist to dive into the hidden resonances and harmonies behind instances in time, while also allowing artists to engage with the history and ecology of the area itself. In my own practice, I have found this level of engagement valuable in constructing narratives within my compositions, and for reflecting on my motivations of creating a piece. These methods can be avenues for reflecting on fieldwork and knowledge gained in the field – for instance, part of *Corridor* is also about acknowledging the local history of the area and the importance of its continual existence.

This style of composition – and more broadly, a more self-reflexive style of working with field recordings – also allows artists a way of drawing attention to current concerns. The Insect Apocalypse series is, at its heart, a series about the increasingly urgent issue of climate change. The sounds in the series are deliberately discordant, haunting and harsh, to highlight the uncomfortable reality that the world is heading towards a crisis.

Ultimately, this approach to field recording – with its emphasis on self-reflexive narrative – can create multi-faceted works that can trigger strong emotional memories and associations in listeners. This method also highlights the active agency of the artist by acknowledging the emotive responses behind the recording (and creation) of a soundscape.

7. REFERENCES

Anderson, I. and Rennie, T. 2016. "Thoughts in the Field: 'Self-reflexive narrative' in field recording". *Organised Sound*, Vol 21, Special Issue 3, pp. 222-232.

Bailie, J. 2014. "Squeezing out the music from real sound." Darmstädter Beiträge zur Neuen Musik. Vol. 22, pp 4-9.

Bailie, J. 2018. "Film Theory and Synchronisation". *CR: The New Centennial Review*, 18:2, pp 69-74.

Cardiff, J. (n.d.) "A Large Slow River." Janet Cardiff & George Bures Miller. https://www.cardiffmiller.com/artworks/walks/large_slow_river.html (accessed September 1, 2019).

English, L. 2017. "A Memory of Almost Nothing: Luc Ferrari's Listening During Presque Rien No. 1". *Leonardo Music Journal*, Vol 27, pp. 17-20.

Jarvis, B. 2018. "The Insect Apocalypse Is Here." *The New York Times Magazine*. https://www.nytimes.com/2018/11/27/magazine/insect-apocalypse.html (accessed September 1, 2019).

Power, S. 2014. "Joanna Bailie: Strange Parallel Music". *Tempo 68*, pp 20-29.

Truax, B. 1992. "Composing with Time-Shifted Environmental Sound". *Leonardo Music Journal* 2:1, pp 37-40.